

What is Amazon EC2

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Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 eliminates your need to invest in hardware up front, so you can develop and deploy applications faster. You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.

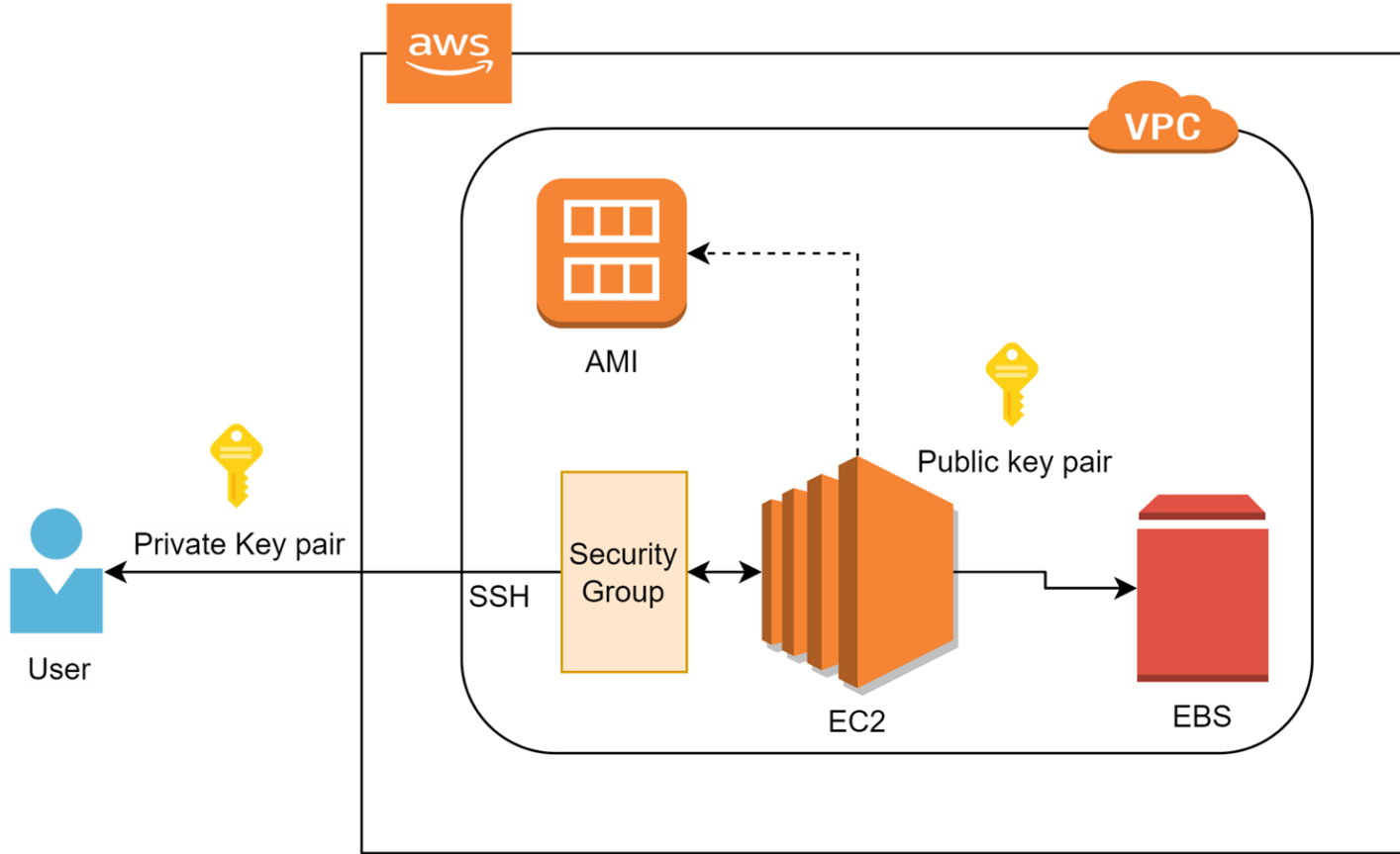


Amazon EC2

Features of Amazon EC2

- Virtual computing environments, known as **instances**
- Preconfigured templates for your instances, known as Amazon Machine Images (AMIs)
- Various configurations of CPU, memory, storage, and networking capacity for your instances, known as **instance types**
- Secure login information for your instances using key pairs (AWS stores the **public key**, and you store the **private key** in a secure place)
- Storage volumes for temporary data that's **deleted** when you stop, hibernate, or terminate your instance, known as instance store volumes
- Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS), known as Amazon EBS volumes
- Multiple physical locations for your resources, such as instances and Amazon EBS volumes, known as **Regions** and **Availability Zones**
- A firewall that enables you to specify the protocols, ports, and source IP ranges that can reach your instances using **security groups**
- Static IPv4 addresses for dynamic cloud computing, known as Elastic IP addresses
- Metadata, known as tags, that you can create and assign to your Amazon EC2 resources
- Virtual networks you can create that are logically isolated from the rest of the AWS Cloud, and that you can optionally connect to your own network, known as virtual private clouds (**VPCs**)
- <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

Launch EC2 Diagram



Launch Windows EC2 Instance

Initiate instance launch [link](#)

Step 1: Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>

Step 2: From the Amazon EC2 console dashboard, choose Launch instance.

Step 3: Choose an Windows Amazon Machine Image (AMI)

Step 4: Choose an Instance Type

Step 5: Configure Instance Details

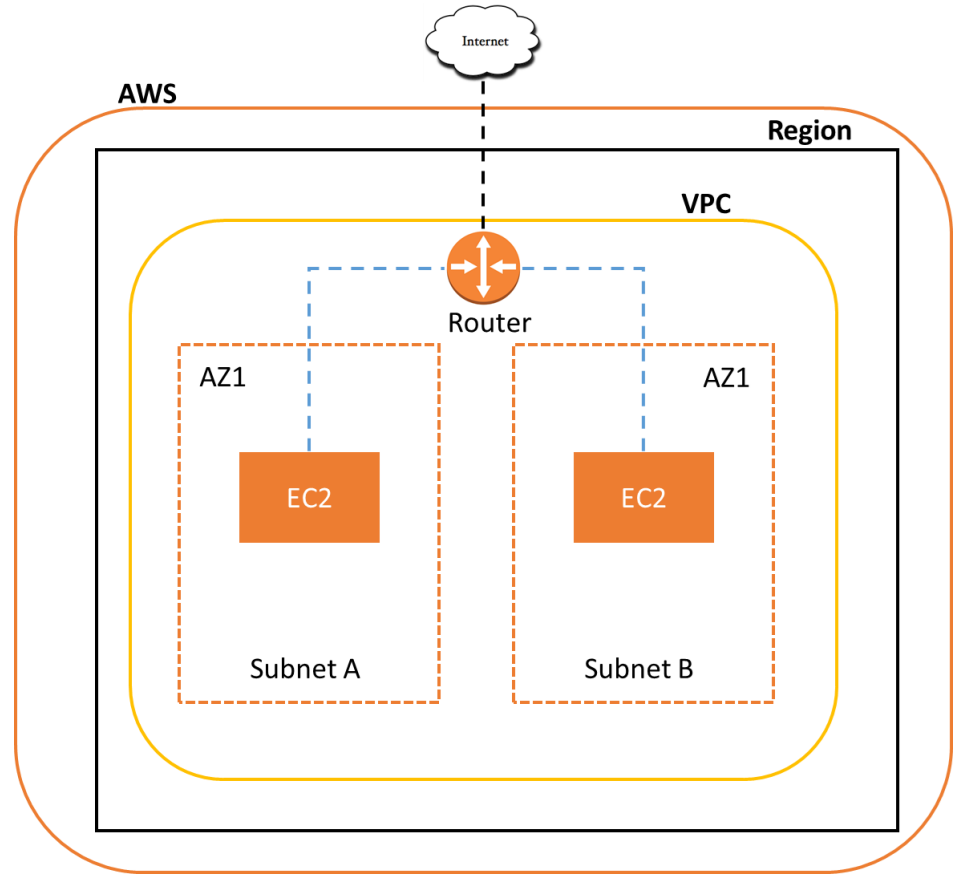
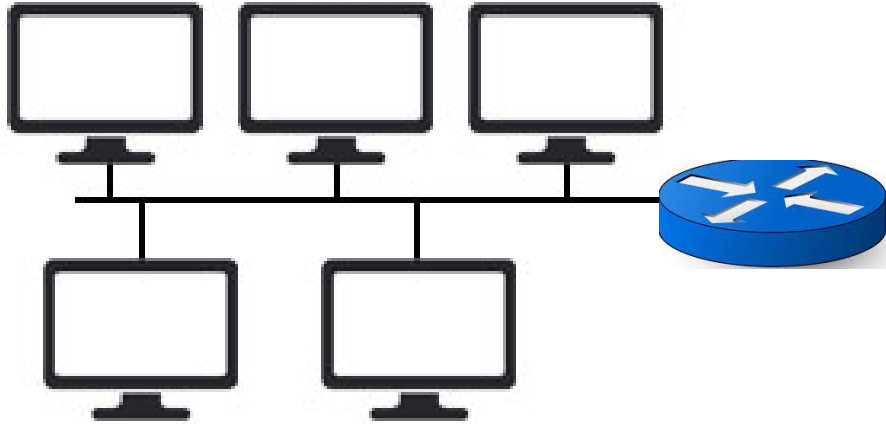
Step 6: Add Storage

Step 7: Add Tags

Step 8: Configure Security Group

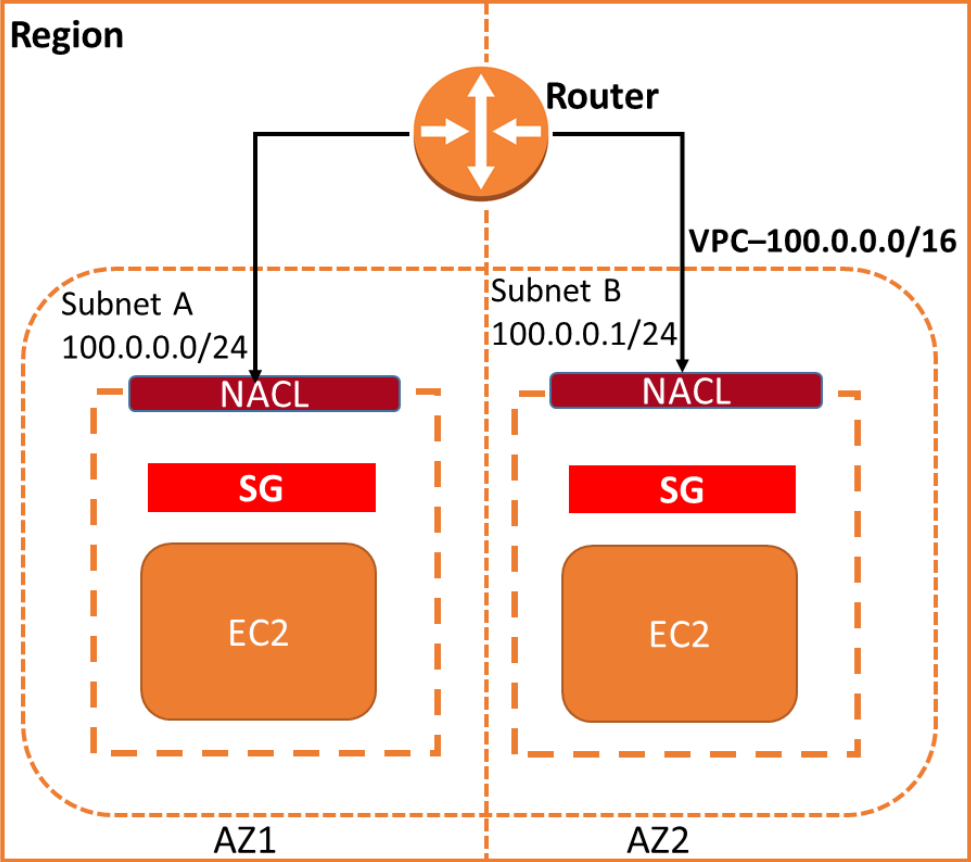
Step 9: Review Instance Launch and Select Key Pair

Traditional Hosting Vs VPC



VPC Concept

- VPC
- Security Group
- NACL
- Route Table-It shows local route
- Subnet
- Availability Zone(AZ)
- Region
- CIDR-100.0.0.0/16



VPC Features

Subnet:

A subnet is a range of IP addresses in your VPC. You can launch AWS resources, such as EC2 instances, into a specific subnet. When you create a subnet, you specify the IPv4 CIDR block for the subnet, which is a subset of the VPC CIDR block.

Subnet Types:

- ❖ **Public Subnet:** The subnet traffic is routed to the public internet through an internet gateway or an egress-only internet gateway.
- ❖ **Private Subnet:** The subnet traffic can't reach the public internet through an internet gateway or egress-only internet gateway. Access to the public internet requires a NAT device.
- ❖ **VPN-only subnet :** The subnet traffic is routed to a Site-to-Site VPN connection through a virtual private gateway. The subnet traffic can't reach the public internet through an internet gateway.

CIDR

Exercise

